Hotel Booking Cancellations

Business Problem Analysis & Strategic Recommendations

City Hotel & Resort Hotel | *Data Period: 2015–2017*

1. Business Problem

City Hotel and Resort Hotel have experienced persistently high booking cancellation rates over recent years. This trend has led to two compounding problems: reduced revenue and suboptimal utilisation of hotel rooms. Reducing cancellation rates is therefore the primary strategic objective for both properties.

This report analyses hotel booking cancellations alongside other factors influencing annual revenue generation, and provides actionable recommendations to address the core business challenge.

2. Assumptions

The following assumptions underpin this analysis:

- No extraordinary events occurred between 2015 and 2017 that would materially distort the dataset.
- The data remains current and is sufficient for evaluating prospective hotel strategies.
- No unforeseen negative consequences are expected from implementing the recommended techniques.
- Neither hotel is currently employing the proposed solutions.
- Booking cancellations are the single largest factor affecting revenue performance.
- A cancellation results in a vacant room for the entire originally booked duration.
- Guests who cancel their reservations typically do so within the same calendar year as the original booking.

3. Research Questions

This analysis seeks to answer three core questions:

- What variables most significantly drive hotel reservation cancellations?
- What targeted actions can reduce the rate of reservation cancellations?
- How can hotels use these insights to inform pricing strategies and promotional campaigns?

4. Hypothesis

Before examining the data, three working hypotheses were established:

- Higher room prices are positively correlated with higher cancellation rates.
- Longer waiting-list periods lead to increased cancellations.
- The majority of reservations originate through offline travel agents rather than direct or online channels.

5. Analysis & Findings

5.1 Overall Reservation Status

Across both properties, a substantial proportion of reservations are successfully completed. However, 37% of all reservations are cancelled — a rate significant enough to materially reduce hotel earnings. This figure establishes the baseline problem the subsequent analysis seeks to explain.

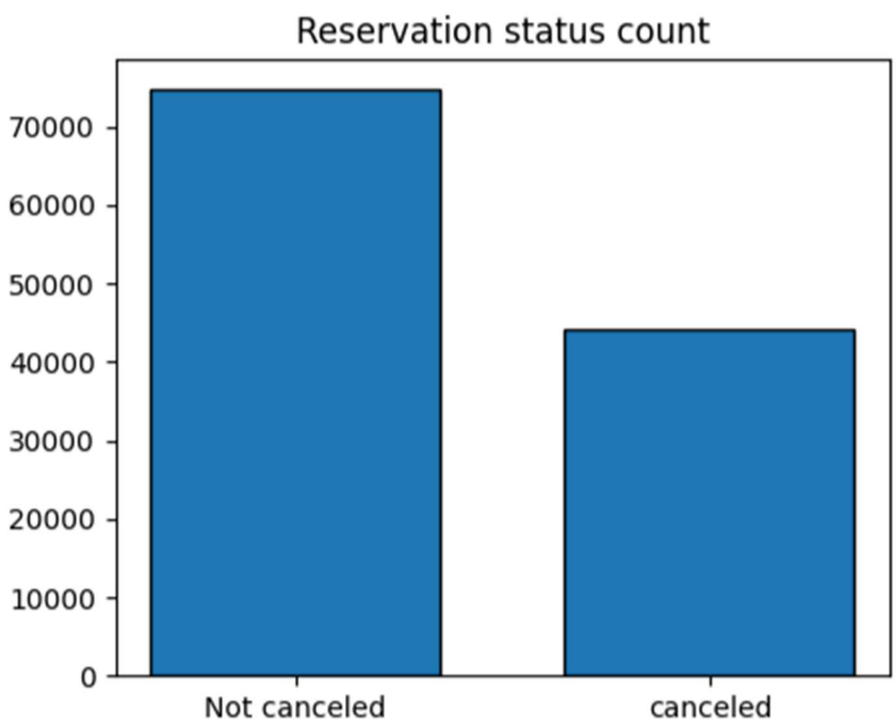

Figure 1: Reservation Status Distribution
Bar chart comparing the share of completed reservations vs. cancellations across both hotels. 37% of reservations are cancelled, representing a significant revenue leakage.

5.2 Booking Volume by Hotel Type

City Hotel consistently records higher booking volumes than Resort Hotel. A likely explanation is that resort properties command premium room rates, making them less accessible for budget-conscious travellers and reducing overall demand relative to city-centre accommodation.

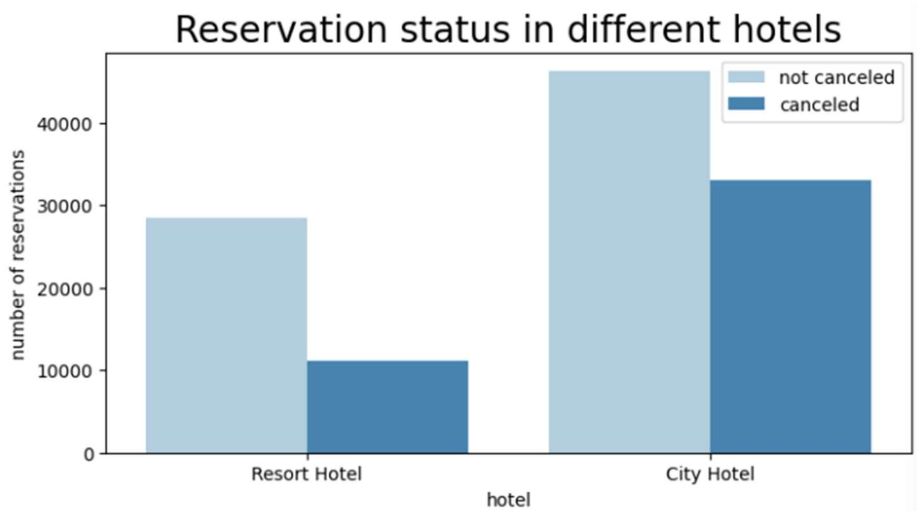

Figure 2: Booking Volume: City Hotel vs. Resort Hotel
Grouped bar chart contrasting total booking volume for City Hotel and Resort Hotel. City Hotel leads significantly, suggesting a price-driven demand differential.

5.3 Average Daily Rate Trends

The line graph tracks the Average Daily Rate (ADR) for both hotel types over time. While city hotels generally maintain a lower ADR, resort hotel rates spike markedly during weekends and public holidays — a pattern consistent with leisure-driven demand surges. This dynamic creates predictable high-risk cancellation windows.

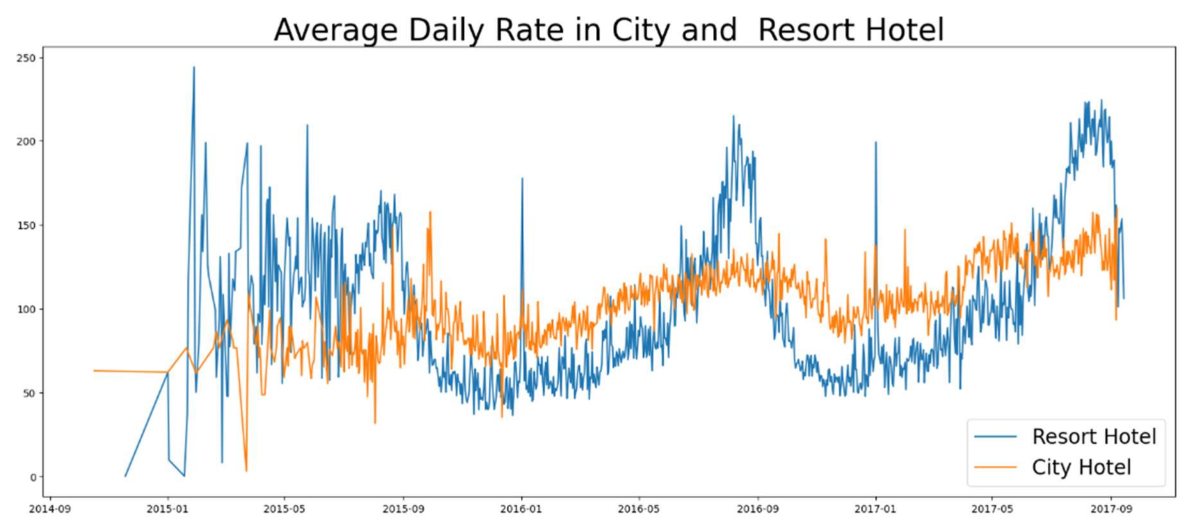

Figure 3: Average Daily Rate (ADR) Over Time — City vs. Resort Hotel
Line graph showing daily ADR fluctuations for both hotel types. Resort Hotel exhibits pronounced weekend and holiday peaks, while City Hotel maintains a more stable, lower rate.

5.4 Monthly Reservation & Cancellation Patterns

Reservation volumes and cancellations both peak in August, indicating that high-demand months also carry the greatest cancellation risk. January stands out as the month with the highest relative cancellation rate, likely reflecting post-holiday budget constraints and a lower overall demand environment.

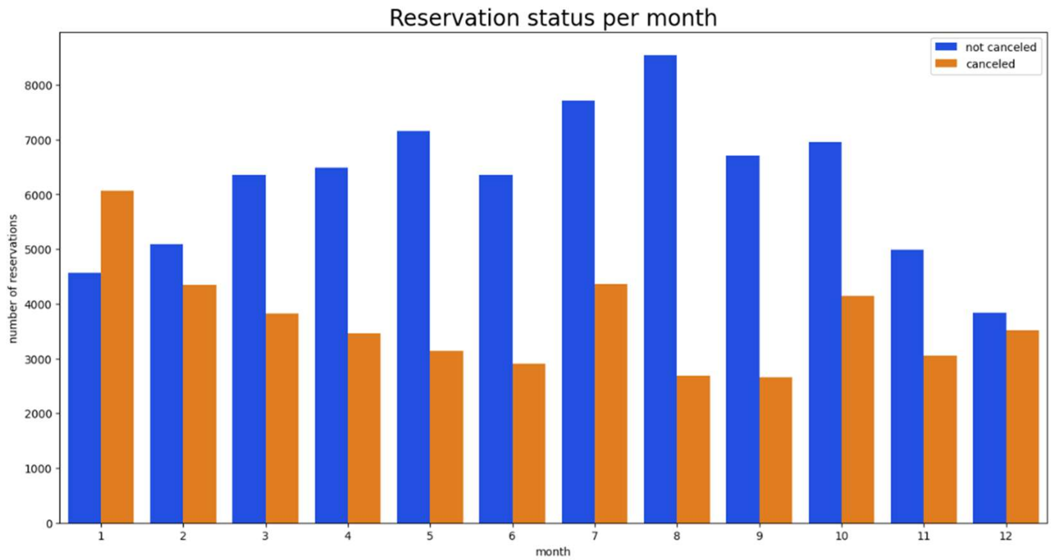

Figure 4: Monthly Reservation Status: Confirmed vs. Cancelled
Grouped bar chart displaying monthly booking and cancellation counts. August has both the highest bookings and highest cancellations; January records the highest cancellation-to-booking ratio.

5.5 Price Sensitivity and Cancellations

A clear positive relationship exists between room price and cancellation rate: cancellations cluster around periods of elevated ADR, while they far less frequent when rates are lower. This validates the first hypothesis and pinpoints pricing as the primary lever for reducing cancellations.

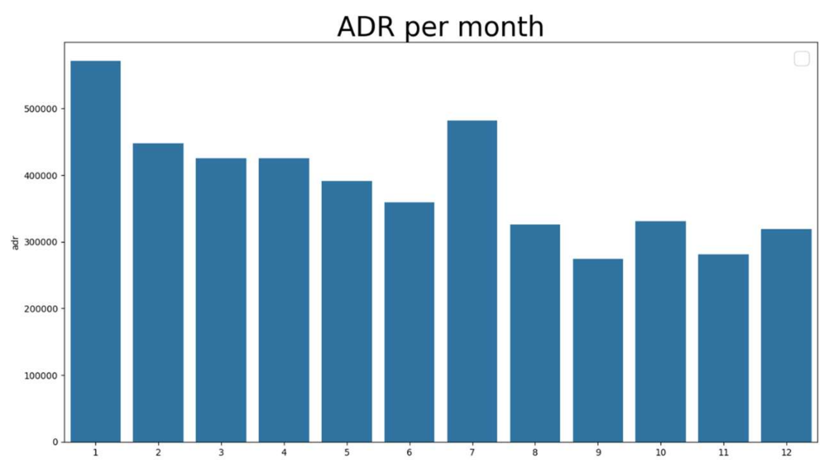

Figure 5: Cancellation Frequency by Price Level
Bar chart showing that cancellation rates are highest when ADR is at its peak and lowest when rates are reduced. Confirms a direct price–cancellation relationship.

5.6 Booking Channel Distribution & Cancellation Rate [Country-wise]

Online travel agencies (OTAs) are by far the dominant booking channel, accounting for approximately 46% of all reservations. Group bookings represent a further 27%. Direct reservations — guests booking by visiting or contacting the hotel — account for only 4% of total volume, highlighting significant over-reliance on intermediary channels.

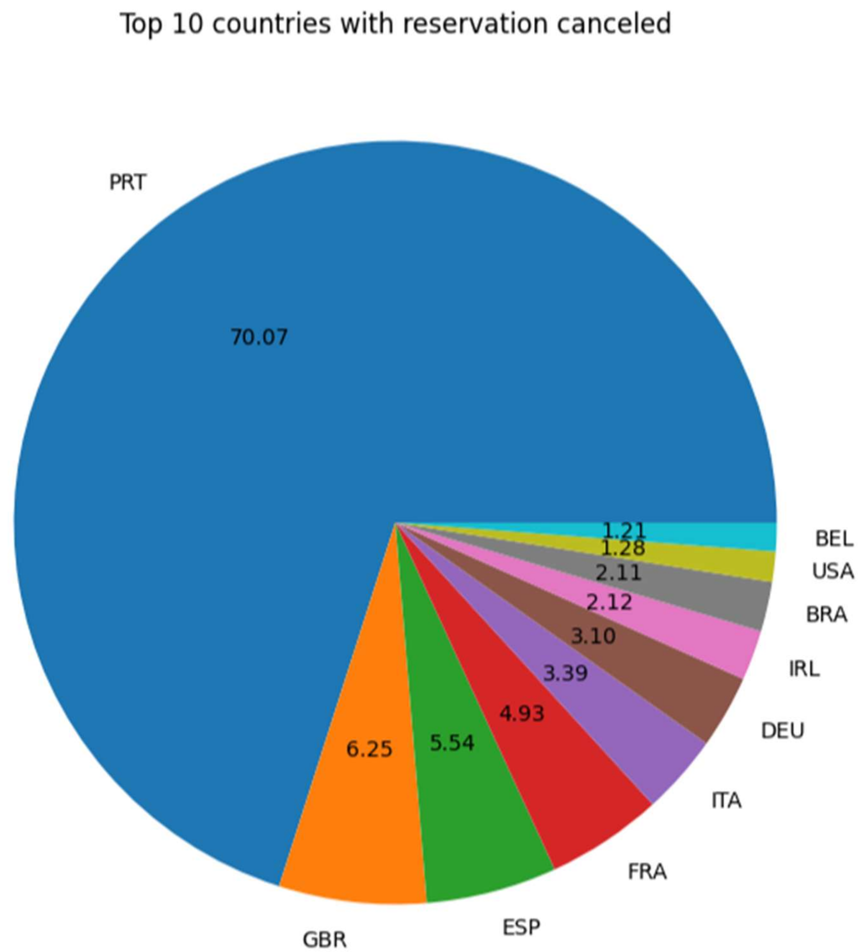

Figure 6: Top 10 countries with maximum cancellation
Pie chart breaking down of the countries with highest cancellation.

Key insight: Portugal records the highest absolute number of cancellations among all source countries, making it a priority market for targeted retention strategies.

5.7 ADR at Cancellation vs. Completion

When comparing the ADR at the time of cancellation against completed stays, cancelled reservations consistently carry a higher average rate. This reinforces the conclusion that price is the decisive factor in cancellation decisions and that rate-management interventions are likely to yield measurable reductions in cancellations.

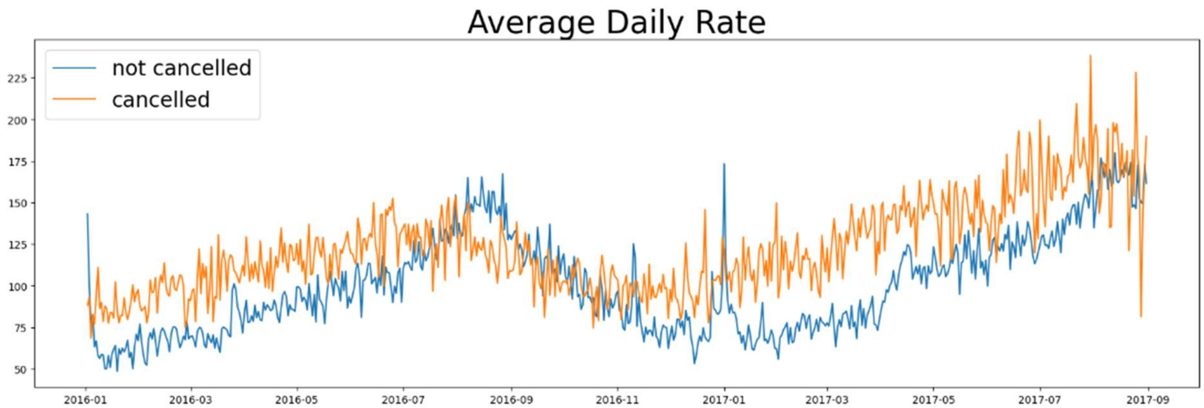

Figure 7: Average Daily Rate: Cancelled vs. Non-Cancelled Reservations
Bar chart contrasting mean ADR for cancelled versus completed reservations. Cancelled bookings have a consistently higher ADR, confirming price as the primary cancellation driver.

6. Strategic Recommendations

Based on the analysis, four targeted recommendations are proposed:

6.1 Dynamic Pricing & Location-Based Discounts

Given the direct link between price and cancellations, both hotels should implement a dynamic pricing model that offers selective discounts during high-ADR periods. Location-specific rate adjustments — particularly for Resort Hotel during holiday peaks — can smooth demand volatility and reduce last-minute cancellations.

6.2 Weekend & Holiday Promotions for Resort Hotel

Resort Hotel exhibits a significantly higher cancellation-to-booking ratio than City Hotel. Introducing weekend and holiday-specific packages with discounted room rates, early-booking incentives, or bundled experiences (e.g., spa, dining) would improve price perception and commitment.

6.3 January Demand Stimulation Campaign

January records the highest relative cancellation rate and lowest booking volumes. A targeted marketing campaign — centred on value-added offers, loyalty rewards, or seasonal packages — launched ahead of this period could meaningfully increase confirmed stays and offset the seasonal revenue dip.

6.4 Portugal Market Retention Strategy

As the country with the highest number of cancellations, Portugal represents a high-priority segment. Both properties should consider service-quality audits, Portuguese-language communications, and loyalty programmes specifically designed to reduce churn within this guest segment.